

AI-Powered NOTAM / METAR / TAF Briefing Assistant

Final (Capstone) Project — General AI Fluency Internship, FlyRank AI

Specialization: Flight Operations / Aviation Tech & AI Integration

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Executive Summary

This capstone presents an AI-powered briefing assistant, built on Claude, that decodes raw NOTAM, METAR, and TAF text into a structured, severity-tiered operational brief for flight dispatch use. The project is grounded in two disciplines I bring together: formal training in flight operations and dispatch (METAR/TAF interpretation, NOTAM analysis, ICAO regulatory compliance, weight and balance basics) and applied AI prompt engineering developed through the FlyRank General AI Fluency Internship.

The assistant is deliberately bounded: it classifies and explains, but it does not calculate fuel or weight and balance, and it is instructed to flag ambiguous or incomplete information rather than guess — a rule drawn directly from aviation safety culture, where visible uncertainty is safer than false confidence.

Five documented test cases anchor the project, each chosen to expose a different limitation of AI-only analysis and the specific human expertise required to close the gap: a partial runway closure (runway-count knowledge), an in-flight TAF amendment (TEMPO-period interpretation), an ILS outage (aircraft equipage knowledge), a wildlife-hazard NOTAM (local seasonal behavior knowledge), and a fuel-unavailable NOTAM at the alternate (regulatory fuel-policy authority). Together, these cases demonstrate that the tool's value lies not in replacing dispatcher judgment, but in accelerating triage while making the boundaries of AI competence explicit and auditable.

The project also situates this design choice within current industry practice, referencing Lufthansa Systems' Lido NOTAM-AI Extractor (December 2025) and The Weather Company's Smart NOTAMs, deployed with Breeze Airways (September 2025) — both of which pair AI-driven summarization with a mandatory human-verification layer, confirming that this capstone's core design principle reflects where the industry is already heading.

1. Project Overview

This project involves designing an AI-based assistant (Claude) capable of analyzing raw NOTAM, METAR, and TAF text and producing a structured, severity-tiered operational brief. The goal is to demonstrate a responsible, human-supervised use of AI in a flight dispatch context, where the dispatcher's human judgment remains the final authority on every operational decision.

This project combines my technical aviation skills (METAR/TAF interpretation, NOTAM analysis, ICAO regulation knowledge) with prompt engineering to produce a tool that is reliable, transparent about its limitations, and useful within a real dispatch workflow.

This capstone is submitted to satisfy the final project requirement of the FlyRank AI General AI Fluency Internship (July 1 – August 31, 2026), applying the program's AI fluency skills to my own field of specialization: flight operations and dispatch.

2. Real-World Problem Addressed

Dispatchers and pilots regularly need to interpret dense, abbreviation-heavy NOTAM and weather text under time pressure. A misread or an overlooked operational constraint (a closed runway, wind shear, reduced visibility) can have direct consequences for flight safety and efficiency.

The proposed assistant speeds up the initial triage of this information while explicitly enforcing one rule: flag uncertainty rather than guess — a core principle of aviation safety culture.

3. Industry Context: Real-World AI Adoption in NOTAM/METAR Processing

This project's approach — using AI to triage and summarize dense operational data while keeping a human verification layer — mirrors a direction the aviation industry itself has been taking in 2025 and 2026. Two concrete examples illustrate this.

Lido NOTAM-AI Extractor — Lufthansa Systems (December 2025)

Lufthansa Systems published an official announcement on December 11, 2025 introducing the Lido NOTAM-AI Extractor, an AI addition to its Lido Aeronautical Data Service. The tool is explicitly positioned as delivering faster, smarter NOTAM processing with human-verified accuracy for global flight operations — the same underlying principle applied in this project's system prompt: AI accelerates interpretation, but a human remains responsible for verifying the result. The product's interface reportedly combines a NOTAM text input, an AI-predicted relevance score, and a map view highlighting the affected airspace, extending plain summarization into a visual operational context.

Smart NOTAMs — The Weather Company, used by Breeze Airways (September 2025)

Smart NOTAMs is a tool developed by The Weather Company (TWC), integrated into its Maverick Dispatch platform, which Breeze Airways adopted and helped shape through direct dispatcher feedback. It was announced in mid-September 2025, with a wider rollout to other TWC products such as Pilotbrief planned for later that year. Rather than presenting a static list, Smart NOTAMs uses AI to detect patterns, summarize meaning, and highlight priority items, grouping notices by relevance such as airport, airspace, or equipment.

A Breeze Airways dispatch manager noted that, on average, only around 15% of the NOTAMs a dispatcher sees actually pertain to their flight, while they may be handling 20 to 30 flights a day — precisely the information-overload problem this capstone's briefing assistant is designed to address. Notably, TWC preserves a human-verification layer similar to this project's design: AI-generated summaries never remove the original NOTAM from the system, each summary links back to its source notice, and the company applies error-catching scoring and user feedback loops to maintain accuracy.

Together, these two industry examples confirm that the core design choice behind this capstone — AI-accelerated triage paired with an explicit, undeletable human verification layer — reflects where major aviation data providers are already heading, rather than an isolated academic exercise.

4. Methodology and Evaluation Approach

The system prompt was not written once and left unchanged — it went through three iterations, each triggered by a specific failure observed during testing.

Iteration 1 — Unstructured Summary

The first version simply asked the model to “summarize the NOTAM and weather.” This produced fluent but operationally unhelpful prose: a single paragraph mixing critical and trivial information with no way for a dispatcher to triage it quickly. This version is preserved as the “BEFORE” example in Section 6.

Iteration 2 — Adding the Severity Tiers

Introducing the CRITICAL / MODERATE / INFORMATIONAL structure immediately improved usability, but testing on the HKNW case revealed a new problem: the model treated the runway closure as a total airport closure, overstating severity because it had no way to know HKNW has four runways. This was not a prompt-wording problem — it was a knowledge gap that no instruction could close, which led directly to Rule 4 (flag rather than guess) and to documenting dispatcher corrections as a first-class part of the deliverable, rather than treating them as edge-case footnotes.

Iteration 3 — Explicit Human-Verification Tier and Scope Limits

The final iteration added the fourth tier (FLAGGED FOR HUMAN VERIFICATION) as a distinct category rather than folding it into MODERATE, and added Rule 5 excluding all calculations from the assistant's authority. This iteration was tested against all five documented cases in this report before being finalized.

Evaluation Criteria

Each test case was evaluated against three questions, applied consistently across all five scenarios:

- Correctness: did the assistant correctly decode the raw text and assign a defensible severity tier?
- Completeness: did it surface every operationally relevant item, without omission?
- Honesty about limits: did it flag what it could not verify, instead of presenting an inference as fact?

This third criterion proved the most diagnostic. In every one of the five cases, the assistant's raw decode was technically accurate — the failure mode was never a misread of the METAR/TAF/NOTAM syntax itself. The value added by dispatcher review was always about context the text does not contain: aircraft equipage, local wildlife behavior, ATC real-time authority, or regulatory fuel policy. This is the central finding of the evaluation and the reason the five cases were chosen to each isolate one such context gap.

5. Risk Assessment and Severity Framework

The four-tier severity system is not arbitrary — each tier maps to a specific operational consequence and a specific action expected of the dispatcher. This section makes that mapping explicit, since it is the backbone every test case in this report is judged against.

CRITICAL

Definition: the item affects runway or navigation-aid availability required for the planned approach, or otherwise creates a direct safety-of-flight constraint. Expected dispatcher action: immediate review before release; the flight cannot be dispatched as originally planned without an explicit decision on this item.

MODERATE

Definition: the item affects operational efficiency or requires a workaround (e.g., a longer taxi route, a reduced payload consideration, a schedule buffer) but does not by itself block dispatch. Expected dispatcher action: factor into the release decision and communicate to the crew, but no automatic hold is warranted.

INFORMATIONAL

Definition: the item has no direct operational impact on this flight but is retained for situational awareness (e.g., confirming an alternate's normal conditions). Expected dispatcher action: no action required beyond awareness.

FLAGGED FOR HUMAN VERIFICATION

Definition: the assistant has identified a point where its confidence is insufficient, or where the answer depends on information categorically outside the NOTAM/METAR/TAF text (aircraft-specific equipage, live ATC coordination, regulatory fuel policy, local seasonal patterns). Expected dispatcher action: mandatory manual verification; this tier can never be silently downgraded or dismissed by the assistant.

This last tier is deliberately treated as structurally different from the first three: CRITICAL, MODERATE, and INFORMATIONAL are AI classifications of what the text says; FLAGGED FOR HUMAN VERIFICATION is an admission of what the text cannot say. Conflating the two would defeat the purpose of Rule 4.

6. System Prompt and Design Rationale

Below is the system prompt used to configure the assistant, followed by an explanation of each rule:

You are an aviation operations assistant specializing in NOTAM and METAR/TAF interpretation for flight dispatch support.

Task: given raw NOTAM and/or METAR/TAF text, produce a structured operational brief.

Rules:

1. Decode all abbreviations and ICAO codes into plain operational language.
2. Categorize each item into exactly one severity tier:
 - CRITICAL: affects runway availability, navigation aids required for approach, or creates a safety-of-flight constraint.
 - MODERATE: affects efficiency or requires an operational workaround but doesn't block dispatch.
 - INFORMATIONAL: no direct operational impact.
3. For each item, state: (a) what it is, (b) the operational impact, (c) the effective time window.
4. If information is ambiguous, incomplete, or you are not confident in the decode, explicitly flag it as "REQUIRES HUMAN VERIFICATION" rather than guessing.
5. Do not perform calculations (fuel, weight & balance) – flag any such requirement for manual dispatcher review instead.
6. Output in the exact format specified. Do not add commentary outside this format.

Rationale for Key Rules

Rule 2 (severity hierarchy): enables rapid triage aligned with a dispatcher's actual operational logic — not just a flat list of information.

Rule 4 (flag rather than guess): the most important rule in the project. It prevents the AI from producing a falsely confident answer on ambiguous information — an unacceptable risk in an aviation safety context.

Rule 5 (no calculations): deliberately bounds the AI's role. Fuel and weight & balance calculations remain a human responsibility, never delegated to the assistant.

7. Documented Test Case 1 — Partial Runway Closure (HKNW–HKMO)

The case below was built to represent a realistic dispatch scenario: a planned flight between HKNW and HKMO, with HKML as the alternate airport, over the 1000Z–1200Z window on July 18, 2026.

Source Data

NOTAM (HKNW):

```
A5678/26 NOTAMN
Q) HKNA/QMRLC/IV/NB0/A/000/999/
A) HKNW B) 2607181000 C) 2607181200
E) RWY 26 CLSD DUE WORK IN PROGRESS
```

METAR (HKNW):

```
HKNW 181000Z 12014G22KT 3500 -RA SCT012 BKN020 21/19 Q1011 NOSIG
```

TAF (HKNW):

```
TAF HKNW 180900Z 1809/1909 11012KT 5000 BKN015
BECMG 1810/1812 13018G26KT 2500 RA BKN008 OVC012
```

METAR (HKMO — destination):

```
HKMO 181000Z 09010KT 6000 SCT018 24/20 Q1012 NOSIG
```

METAR (HKML — alternate):

```
HKML 181000Z 08008KT 8000 FEW020 25/19 Q1013 NOSIG
```

8. Before / After Comparison (Demonstrating the Value of Prompt Engineering)

BEFORE — Basic Prompt ("Summarize this NOTAM and weather for me")

Summary produced by a generic prompt, with no structure and no caution rules:

"There is a NOTAM at HKNW indicating the runway is closed for maintenance, effective from 1000Z to 1200Z on July 18. Current weather at HKNW shows moderate rain with a broken cloud layer at 1200ft, wind 120° at 14kt gusting 22kt, and visibility of 3500m. The TAF suggests a slight deterioration in conditions during the period. Nearby airports HKMO and HKML report clear conditions and remain suitable diversion options."

Weaknesses of this version:

- Treats "RWY 26 CLSD" as if the entire airport were closed, without accounting for the fact that HKNW has 4 runways.

- Does not flag that the NOTAM does not mention real-time runway assignment, which falls under ATC authority, not the NOTAM text itself.
- Buries the overlap between the partial runway closure and the weather deterioration in a vague sentence, without surfacing it as a distinct point of attention.

AFTER — With the Engineered System Prompt

CRITICAL

RWY 26 CLSD (maintenance): Runway 26 is closed for maintenance at HKNW. The NOTAM does not explicitly state real-time runway assignment because this falls under ATC authority, not the NOTAM text. | Effective: 1000Z–1200Z (18/07).

MODERATE

Wind direction vs. closed runway: forecast wind 130°/18–26kt per the TAF (BECMG period). Verify whether runway 26 was the preferred runway given this wind direction — if so, its closure would increase load on the remaining runways during a period of strong wind.

Surface wind at HKNW: 120°/14kt gusting 22kt — within normal limits.

INFORMATIONAL

HKMO (destination): clear conditions, NOSIG, no operational impact expected.

HKML (alternate): clear conditions, NOSIG, remains a viable diversion option.

FLAGGED FOR HUMAN VERIFICATION

ATC confirmation required: final coordination of remaining runway usage rests with local ATC. A dispatcher must confirm via ATC or ATIS the actual runway configuration in use during the 1000Z–1200Z window, rather than relying solely on the NOTAM text.

Overlap between partial closure and weather deterioration: confirm whether the closed runway corresponds to the runway normally used given the forecast wind — an unfavorable runway orientation would further reduce operational capacity.

9. Documented Test Case 2 — In-Flight TAF Amendment (HKJK–HKMO)

This second case was added to broaden the demonstrated scope of the assistant beyond a static NOTAM closure. It tests a different failure mode: a TAF amendment (TAF AMD) published mid-flight-window, requiring the assistant to detect a temporal discrepancy between an initial forecast and a real-time update, rather than simply decoding a single static document.

Scenario

Route Nairobi (HKJK) → Mombasa (HKMO), alternate Malindi (HKML), planned flight window 0600Z–0800Z on the same operational day. An initial TAF was issued at 0000Z; a TAF amendment was subsequently issued at 0645Z — after the flight window had already begun — reflecting a rapid deterioration in convective weather at the destination.

Source Data

TAF initial (issued 0000Z, valid 0100/0224):

```
TAF HKMO 010000Z 0100/0224 09012KT 9999 FEW020  
BECMG 0106/0108 21008KT
```

TAF amendment (issued 0645Z, mid-window):

```
TAF AMD HKMO 010645Z 0106/0212 24018G28KT 3000 TSRA SCT015CB BKN025  
TEMPO 0106/0109 1500 +TSRA BKN008CB
```

NOTAM (alternate HKML — operating hours, unchanged):

```
A1234/26 NOTAMN  
Q) HKNA/QFAAH/IV/NB0/A /000/999/0316S03958E005  
A) HKML B) 2607010000 C) 2607312359  
E) AD OPS HR SKED 0400-1600Z
```

Assistant Output — With the Engineered System Prompt

CRITICAL

TAF amendment at HKMO: destination weather has deteriorated significantly from the original forecast — thunderstorms (TSRA) with cumulonimbus cloud (SCT015CB, BKN025) and visibility reduced to 3000m, versus the original forecast of 9999m visibility and FEW020. This directly affects approach and landing safety margins at the destination. | Effective: 0106Z onward (amendment issued 0645Z, i.e., after the flight window opened at 0600Z).

MODERATE

Temporal discrepancy flagged: the amendment was published 45 minutes after the flight window began. Any briefing generated before 0645Z using only the initial TAF would be stale for the back half of the window; the dispatcher should confirm which TAF version was used for the original release decision.

TEMPO period distinction: the most severe conditions (visibility 1500m, +TSRA, BKN008CB) apply only to the TEMPO sub-period 0106/0109Z, not to the full 0106/0212Z validity of the amendment. The assistant must not represent the entire amended period as under continuous severe thunderstorm conditions.

INFORMATIONAL

HKML (alternate): NOTAM confirms scheduled operating hours 0400–1600Z, which covers the 0600Z–0800Z flight window — no scheduling conflict identified from this NOTAM alone.

FLAGGED FOR HUMAN VERIFICATION

Re-dispatch or hold decision: the assistant identifies the deterioration and the CRITICAL tier but does not recommend delaying, re-routing, or holding the flight. That operational decision rests entirely with the dispatcher, consistent with the project's core design principle.

Alternate suitability under convective weather: confirm that HKML is not itself affected by the same convective system before relying on it as the alternate, since this amendment only covers HKMO.

Dispatcher Review and Correction — Case 2

A generic or less carefully specified AI output could plausibly have applied the TEMPO conditions (1500m, +TSRA, BKN008CB) across the entire amended validity period (0106/0212Z), overstating the severity and duration of the event. As a dispatcher, I know that a TEMPO group describes a temporary, non-continuous condition expected to occur for less than half of the indicated sub-period — here, only 0106/0109Z. This is the same category of correction as the runway-count issue in Case 1: a piece of domain knowledge that the NOTAM/TAF text does not spell out, and that no prompt alone can substitute for. It confirms that the assistant's output tier (CRITICAL, correctly assigned due to the underlying TSRA/CB and reduced visibility outside the TEMPO window as well) is sound, while its duration framing requires a dispatcher's read to be used correctly.

10. Documented Test Case 3 — ILS Outage at Destination (HKJK–HKKI)

This case tests a failure mode centered on aircraft equipage rather than weather or airport infrastructure alone: an unserviceable ILS combined with marginal weather at the destination, where the correct response depends on information about the specific aircraft that no NOTAM, METAR, or TAF can ever contain.

Scenario

Route Nairobi (HKJK) → Kisumu (HKKI), planned arrival 0500Z. The ILS for the only precision-capable runway at HKKI is unserviceable, and the destination is reporting a ceiling and visibility close to the minima for the remaining non-precision approach.

Source Data

NOTAM (HKKI):

```
B2211/26 NOTAMN
Q) HKNA/NAVSHU/IV/NB0/A/000/999/
A) HKKI B) 2607200300 C) 2608202359 EST
E) ILS RWY 13 U/S
```

METAR (HKKI):

```
HKKI 200500Z 05006KT 2400 BR BKN004 OVC008 18/17 Q1015 NOSIG
```

Assistant Output — With the Engineered System Prompt

CRITICAL

ILS RWY 13 unserviceable at HKKI: precision approach is unavailable at the destination for the estimated NOTAM period 2607200300–2608202359. Current METAR shows a 400ft ceiling and 2400m visibility in mist — conditions close to typical non-precision-approach minima. This combination directly affects the ability to land as planned. | Effective: from 0300Z on 20/07, estimated through 20/08.

MODERATE

NOTAM validity marked EST (estimated): the end date is not firm and should be reconfirmed closer to the operation, as ILS restoration timelines commonly shift.

INFORMATIONAL

Wind and temperature at HKKI are otherwise unremarkable and do not independently constrain the approach.

FLAGGED FOR HUMAN VERIFICATION

Aircraft approach capability: whether the planned aircraft and crew are approved for the applicable non-precision or RNP approach at the minima required by current weather is outside the scope of the NOTAM/METAR/TAF text and cannot be determined by this assistant.

Dispatcher Review and Correction — Case 3

This is the clearest illustration in the report of a limit the assistant cannot close on its own, even with a perfect decode. Whether the flight can still be dispatched depends on the specific aircraft's navigation equipage (RNP AR approval, GPS/WAAS capability) and the crew's approach qualifications — information that lives in the operator's ops specs and the aircraft's equipment list, not in any aeronautical information product. As a dispatcher, this is precisely the kind of check I would run before even looking at the weather trend, and it is the correct instinct behind Rule 4: the assistant does the right thing by refusing to assume equipage rather than defaulting to either an optimistic or pessimistic guess.

11. Documented Test Case 4 — Wildlife Hazard NOTAM (HKMO)

This case tests a different kind of gap: one where the NOTAM text is unambiguous, but assigning the correct operational weight to it depends on local, seasonal knowledge that the text does not — and structurally cannot — provide.

Scenario

Route arriving at Mombasa (HKMO) at 1530Z, during the late-afternoon period when bird activity near the coast is known locally to increase.

Source Data

NOTAM (HKMO):

```
C0987/26 NOTAMN
Q) HKNA/OLXXX/IV/NBO/A/000/200/
A) HKMO B) 2607150000 C) 2608150000 EST
E) BIRD CONCENTRATION VICINITY RWY 03/21 THR
CAUTION ADVISED
```

Assistant Output — With the Engineered System Prompt

MODERATE

Bird concentration reported near both runway thresholds at HKMO for an extended, estimated period (15 July–15 August). This does not by itself prevent dispatch but warrants crew briefing on bird-strike caution during approach and departure. | Effective: 2607150000–2608150000 (estimated).

INFORMATIONAL

No associated runway or navigation-aid restriction is stated in this NOTAM; it is an advisory only.

FLAGGED FOR HUMAN VERIFICATION

Time-of-day risk variation: whether this specific arrival time (1530Z) falls within a period of locally elevated bird activity is not stated in the NOTAM and cannot be assessed from this text alone.

Dispatcher Review and Correction — Case 4

A generic AI reading could reasonably leave this at MODERATE and move on, since the NOTAM text itself gives no reason to escalate. As a dispatcher operating this route, I know that bird activity at this coastal location is materially higher during the late-afternoon feeding period than the NOTAM's blanket estimated date range alone would suggest — a seasonal and time-of-day pattern that comes from local operational experience, not from any published document. The correct action is not for the AI to invent this knowledge, but for it to have already flagged the time-of-day dimension as unverifiable from the text — which the engineered prompt did. This case shows the same principle as Case 3 in a very different domain: local/behavioral knowledge, not just technical equipment.

12. Documented Test Case 5 — Fuel Unavailable at Alternate (HKML)

This case reinforces the project's Rule 5 boundary (no calculations) by testing a scenario where the natural next question — is the remaining fuel policy still compliant — is one the assistant must explicitly decline to answer.

Scenario

HKML is planned as the alternate for a flight where fuel uplift at the destination is limited; a NOTAM reports the fuel facility at HKML as temporarily unavailable.

Source Data

NOTAM (HKML):

```
D0456/26 NOTAMN
Q) HKNA/FAXXX/IV/NB0/A/000/999/0316S03958E005
A) HKML B) 2607220600 C) 2607230600
E) FUEL JET A1 NOT AVBL DUE SUPPLY DISRUPTION
```

Assistant Output — With the Engineered System Prompt

CRITICAL

Jet A1 fuel is not available at HKML, the planned alternate, for the period 2607220600–2607230600. If HKML is required as a genuine diversion during this window, no fuel uplift will be possible there. | Effective: 0600Z 22/07 – 0600Z 23/07.

FLAGGED FOR HUMAN VERIFICATION

Regulatory fuel compliance: whether the flight's planned fuel load still satisfies applicable minimum alternate-fuel requirements given this restriction is a calculation and regulatory-compliance question that this assistant is explicitly not authorized to perform (Rule 5). This must be assessed manually by the dispatcher against the applicable operations specifications.

Second alternate requirement: whether policy requires nominating a second alternate as a result of this NOTAM is likewise a dispatcher-level policy decision, not a text-decoding task.

Dispatcher Review and Correction — Case 5

This case does not require a factual correction to the assistant's decode — the CRITICAL classification and the two FLAGGED items are both appropriate on their own terms. What it demonstrates instead is the discipline of the scope boundary itself: it would have been easy to design a more “helpful” prompt that attempts a rough fuel-margin estimate, and such an estimate might even often be correct. As a dispatcher, I would treat that kind of unauthorized estimate as more dangerous than no estimate at all, because it invites reliance on a number that was never actually computed against the operator's real fuel policy. Rule 5 exists specifically to prevent this failure mode, and this case is the clearest demonstration of why that boundary was worth setting from the start.

13. Dispatcher Review and Corrections — Summary Across All Cases

This section consolidates the points where dispatch expertise corrected or enriched the AI's initial output across all five documented cases — a central element in demonstrating an “AI with human review” use case rather than blind automation.

Correction 1 — Number of Runways at HKNW (Case 1)

The generic AI output implicitly assumed a total airport closure. As a dispatcher, I know HKNW has 4 runways: the closure therefore only affects runway 26, which significantly changes the assessment of the real operational impact.

Correction 2 — ATC Authority Over Runway Assignment (Case 1)

The NOTAM does not specify real-time runway assignment because that decision falls under ATC, not the NOTAM text. This is a structural limitation of NOTAMs themselves: they inform that a resource is unavailable, but they never replace real-time ATC coordination. No prompt, however well designed, can replace this operational knowledge — it comes directly from the dispatcher's professional judgment.

Correction 3 — TEMPO Period Scope (Case 2)

A TEMPO group describes a temporary, non-continuous condition applying to less than half of its stated sub-period, not the full amendment validity. Recognizing this prevents overstating the duration of a severe-weather event.

Correction 4 — Aircraft Equipage Is Outside Text Scope (Case 3)

Approach capability under a navigation-aid outage depends on the specific aircraft's equipment and crew qualifications — information that no NOTAM, METAR, or TAF can contain, and that the assistant correctly refuses to assume.

Correction 5 — Local Seasonal/Behavioral Knowledge (Case 4)

A wildlife-hazard NOTAM's blanket date range does not capture finer time-of-day risk patterns that come from route-specific operational experience rather than published text.

Correction 6 — Regulatory Fuel Policy Is a Human Decision (Case 5)

Whether a fuel restriction at an alternate still satisfies minimum fuel policy is a regulatory compliance question requiring a calculation the assistant is deliberately not authorized to perform.

Read together, these six corrections fall into three distinct categories: factual/structural knowledge the text omits (Cases 1 and 2), capability knowledge external to the text by design (Case 3), and policy/regulatory authority deliberately reserved for humans (Case 5), with Case 4 sitting between the first two categories. This taxonomy is, in itself, one of the more useful outputs of the project: it gives a reusable framework for deciding, for any future NOTAM/METAR/TAF scenario, which kind of human check is actually required.

14. Glossary of Abbreviations

The following abbreviations appear in the NOTAM, METAR, and TAF examples used throughout this report.

NOTAM	Notice to Air Missions — a notice containing information on the establishment, condition, or change of any aeronautical facility, service, procedure, or hazard.
METAR	Meteorological Aerodrome Report — a routine, scheduled surface weather observation.
TAF	Terminal Aerodrome Forecast — a forecast of expected weather conditions at an airport over a defined period.
TAF AMD	Amended TAF — a revision to a previously issued forecast, published when actual or expected conditions deviate significantly.
RWY	Runway.
CLSD	Closed.
U/S	Unserviceable (out of service).
ILS	Instrument Landing System — a precision approach and landing aid using radio signals.
RNP / RNP AR	Required Navigation Performance / Authorization Required — a satellite-based navigation approach standard requiring specific aircraft and crew approval.
BECMG	Becoming — indicates a gradual, permanent change expected within the stated period.
TEMPO	Temporary — indicates a temporary fluctuation expected to occur for less than half of the stated period.
TSRA	Thunderstorm with rain.
CB	Cumulonimbus cloud — associated with thunderstorm activity.

SCT / BKN / OVC / FEW	Cloud cover amount: Scattered, Broken, Overcast, Few (in ascending order of sky coverage).
BR	Mist (reduced visibility due to fine water droplets).
NOSIG	No significant change expected.
Q1011 / Q1012 / Q1013	QNH — altimeter setting in hectopascals, used to set the aircraft altimeter to reflect airport elevation.
EST	Estimated — indicates the stated end date/time of a NOTAM is provisional.
PPR	Prior Permission Required — an operational restriction requiring advance authorization before arrival.
ATC	Air Traffic Control.
ATIS	Automatic Terminal Information Service — a continuous broadcast of routine airport and weather information.
ICAO	International Civil Aviation Organization — the UN agency setting global standards for civil aviation.

15. Limitations and Scope of the Project

This tool is a portfolio/training prototype and is not certified for real operational dispatch use. It is designed to demonstrate an AI-assisted interpretation workflow under human supervision, not to replace a certified dispatch system or a licensed dispatcher's judgment.

- No calculation authority: the AI never performs fuel, weight & balance, or performance calculations — these remain entirely manual.
- Uncertainty is flagged, not resolved: in cases of ambiguous phrasing, the model is instructed to flag the item for human verification rather than infer a likely meaning.
- No real-time data validation: the tool processes whatever text it is given, without independently verifying the freshness or authenticity of the NOTAM/METAR/TAF source.
- The severity hierarchy reflects general ICAO operational norms, not a specific airline's operations manual.
- Human review remains the final authority on every output tier, including CRITICAL items.

What I Learned

The most important design decision in this project was not the summarization itself, but the explicit instruction “don't guess, flag it.” Early testing without this instruction produced severity classifications that were sometimes wrong but stated with misleading confidence on ambiguous inputs. Aviation safety culture treats false confidence as more dangerous than visible uncertainty — a principle that should guide the design of AI tools in this domain.

Extending the assistant to a second, temporally dynamic scenario (Case 2) reinforced this lesson further: correctly scoping a TEMPO period versus a continuous condition is exactly the kind of nuance where an AI tool must either get it precisely right or explicitly flag it — there is no safe middle ground in an aviation context.

16. Future Improvements and Roadmap

This capstone represents a first, deliberately bounded version of the assistant. The following extensions were identified during the project but intentionally left out of the current scope, either due to time constraints within the internship or because they would require infrastructure beyond a single-model prompt design.

- Structured JSON output mode: alongside the human-readable brief, produce a machine-parseable JSON version of the same tiers, enabling integration into a dispatch dashboard rather than a chat-only workflow.
- Aircraft-equipage lookup: connect the assistant to a (hypothetical) fleet-equipage reference so that Case 3-style questions about RNP/ILS capability could be answered directly rather than only flagged — this would need to be built and validated as a separate, auditable data source, not inferred by the model.
- Multi-airport route batching: extend the assistant to process an entire route (departure, destination, and multiple alternates) in a single structured pass, rather than one airport's NOTAM/METAR/TAF set at a time.
- Historical NOTAM pattern flags: for recurring advisory NOTAMs such as wildlife hazards (Case 4), track how often a given NOTAM has been reissued to help distinguish routine seasonal advisories from newly emerging risks — without ever asking the model to guess at behavior the text does not state.
- Formal accuracy benchmarking: evaluate the assistant against a larger, independently-reviewed set of real (anonymized) NOTAM/METAR/TAF combinations, scored by working dispatchers, to move from five illustrative cases to a statistically meaningful accuracy measure.

Automation platforms such as Make.com and Zapier were evaluated early in the project for orchestrating this workflow and deliberately set aside in favor of a direct, single-model Claude Project workflow, given the internship's learning objectives and the added complexity such platforms would introduce without a corresponding benefit at this stage. This remains a candidate to revisit only once the assistant has an actual multi-step, multi-system workflow to orchestrate — which the roadmap above would create.

17. References and Bibliography

- International Civil Aviation Organization (ICAO), Annex 15 to the Convention on International Civil Aviation — Aeronautical Information Services.
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- Lufthansa Systems, Lido NOTAM-AI Extractor product announcement, December 2025.
- The Weather Company, Smart NOTAMs feature within the Maverick Dispatch platform, announced September 2025; deployment referenced with Breeze Airways.

- Kenya Aeronautical College, Wilson Airport, Nairobi — Flight Operations and Dispatch program curriculum, foundational reference for METAR/TAF interpretation, NOTAM analysis, and ICAO regulatory compliance training applied in this project.
- FlyRank Corp. (FlyRank AI), General AI Fluency Internship curriculum, July–August 2026 cohort.

18. Conclusion

This project demonstrates a responsible, human-supervised use of artificial intelligence in a flight dispatch context: the AI accelerates the triage and synthesis of dense information, while the dispatcher's professional judgment remains the final authority on every operational decision. The before/after comparison and the five documented test cases — a static partial runway closure, a dynamic in-flight TAF amendment, an ILS outage bound by aircraft equipage, a wildlife-hazard NOTAM bound by local seasonal knowledge, and a fuel-unavailable NOTAM bound by regulatory policy — concretely illustrate the added value of prompt design and human expertise across a deliberately varied set of operational scenarios, beyond the model's raw capability alone.

The methodology and risk-assessment sections of this report show that this was not incidental: each case was chosen to isolate a specific category of limitation, and the resulting taxonomy of dispatcher corrections is itself a transferable framework for evaluating any future AI-assisted dispatch tool, not just this one.